

C-12

Prévoir le comportement fréquentiel des SLCI

Travaux Dirigés

2020-2021

TD 3 : Comportement fréquentiel des SLCI, intégrateur, dérivateur et 1° ordre

Exercice 3.1 : Tracer de diagramme de Bode

On donne les fonctions de transfert suivantes :

$$H_1(p) = \frac{5}{1 + \frac{p}{10}}$$

$$H_4(p) = 1 + \frac{p}{50}$$

$$H_7(p) = \frac{12}{p} \times \frac{1}{\left(1 + \frac{p}{10}\right)\left(1 + \frac{p}{1000}\right)}$$

$$H_2(p) = \frac{10}{1 + \frac{p}{5}}$$

$$H_5(p) = 1 + \frac{p}{7}$$

$$H_8(p) = \frac{\frac{p}{10}}{1 + \frac{p}{10}}$$

$$H_3(p) = \frac{1}{1 + \frac{p}{100}}$$

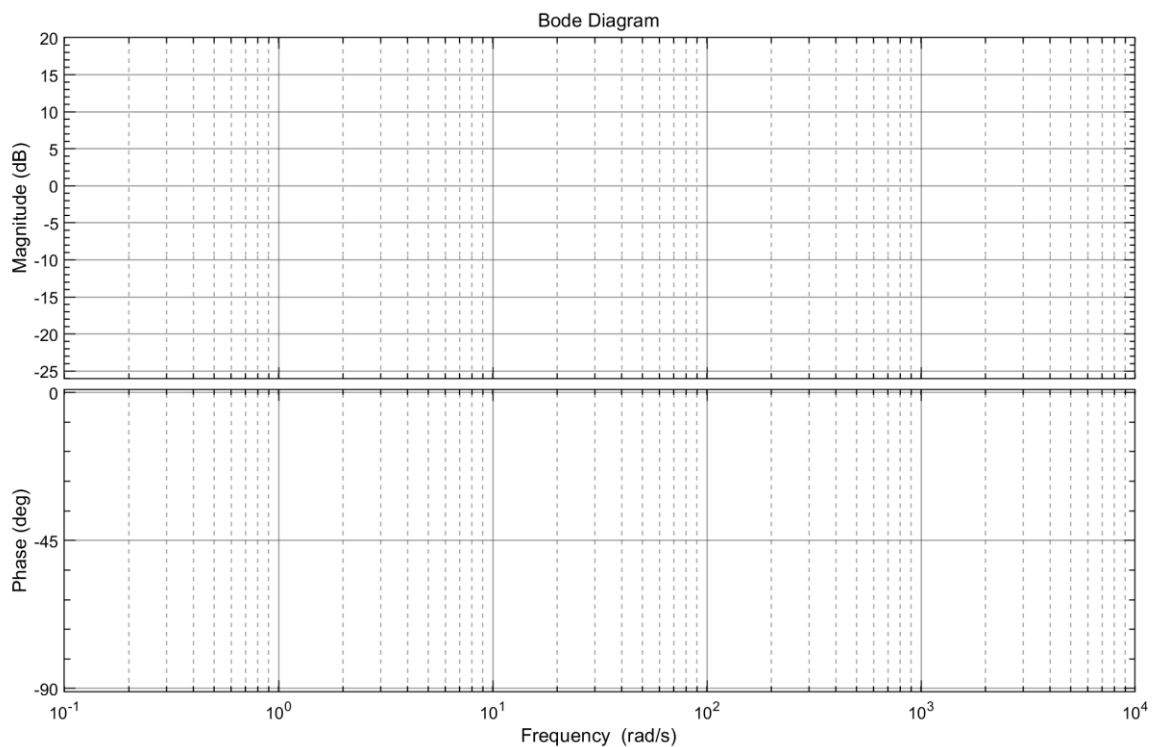
$$H_6(p) = \frac{5\left(1 + \frac{p}{1000}\right)}{1 + \frac{p}{10}}$$

$$H_9(p) = 14 \times \frac{\frac{p}{5}}{1 + \frac{p}{5}}$$

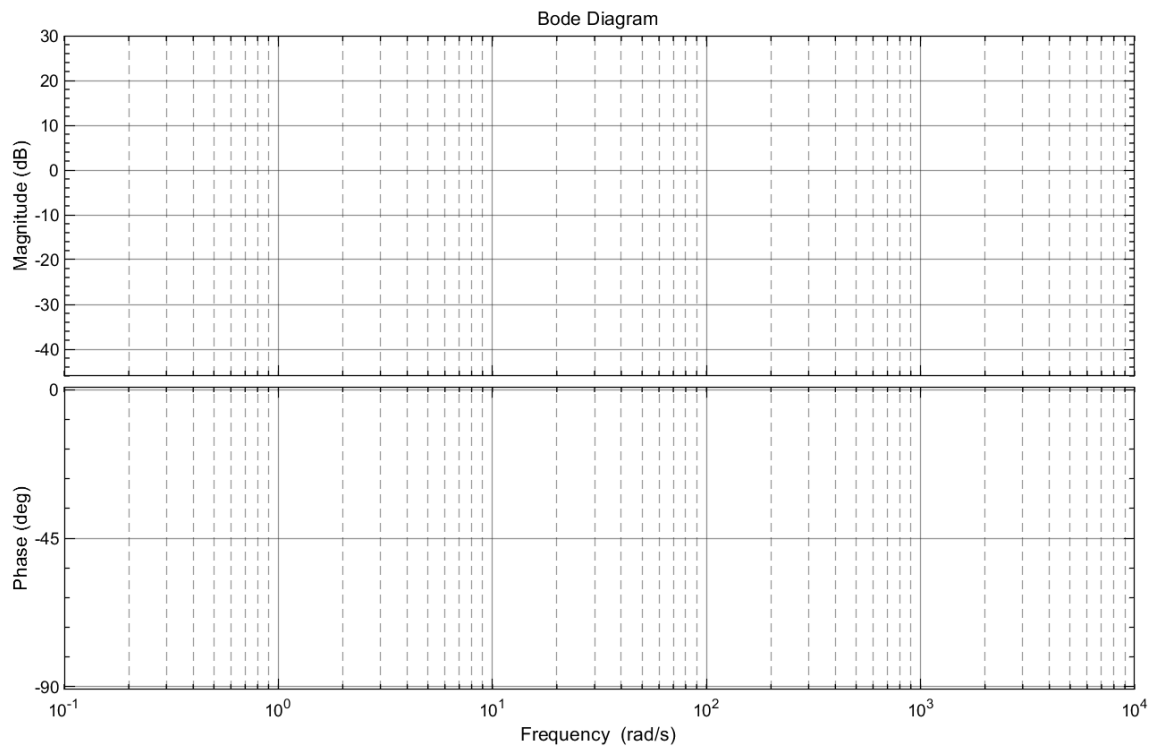
Tracer les diagrammes de Bode pour ces fonctions de transferts.

Le numéro de chaque diagramme est directement lié à l'indice de la fonction de transfert.

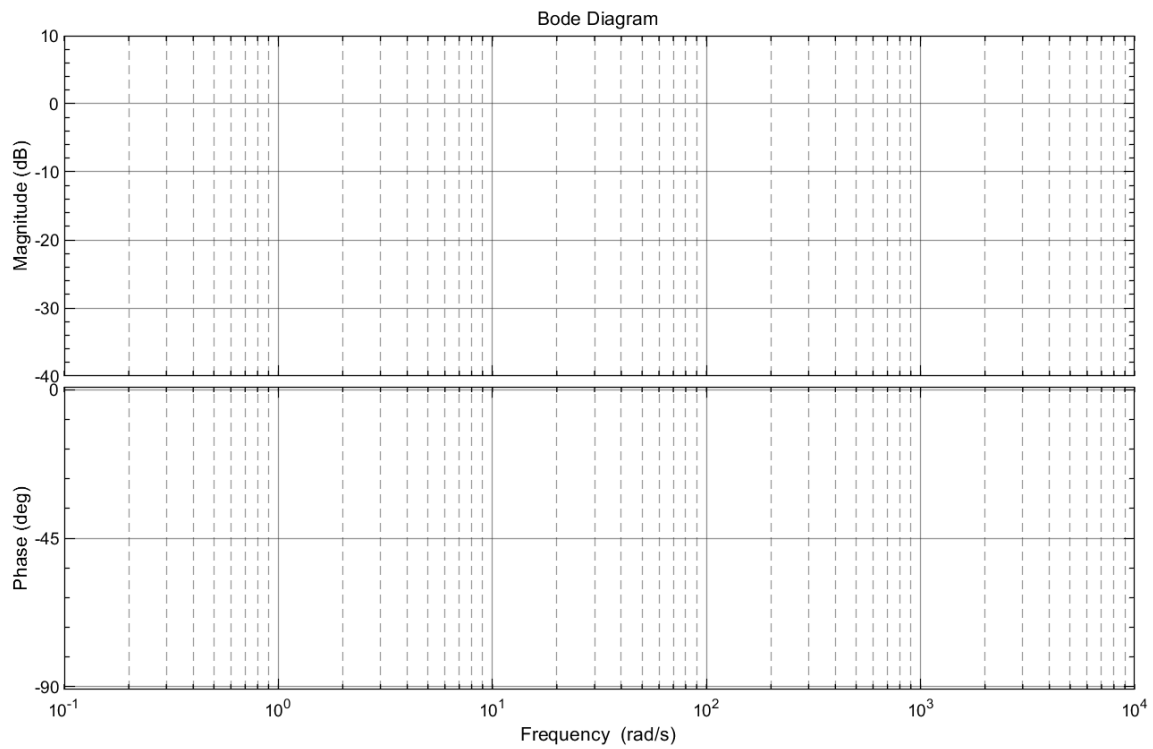
Q1 :



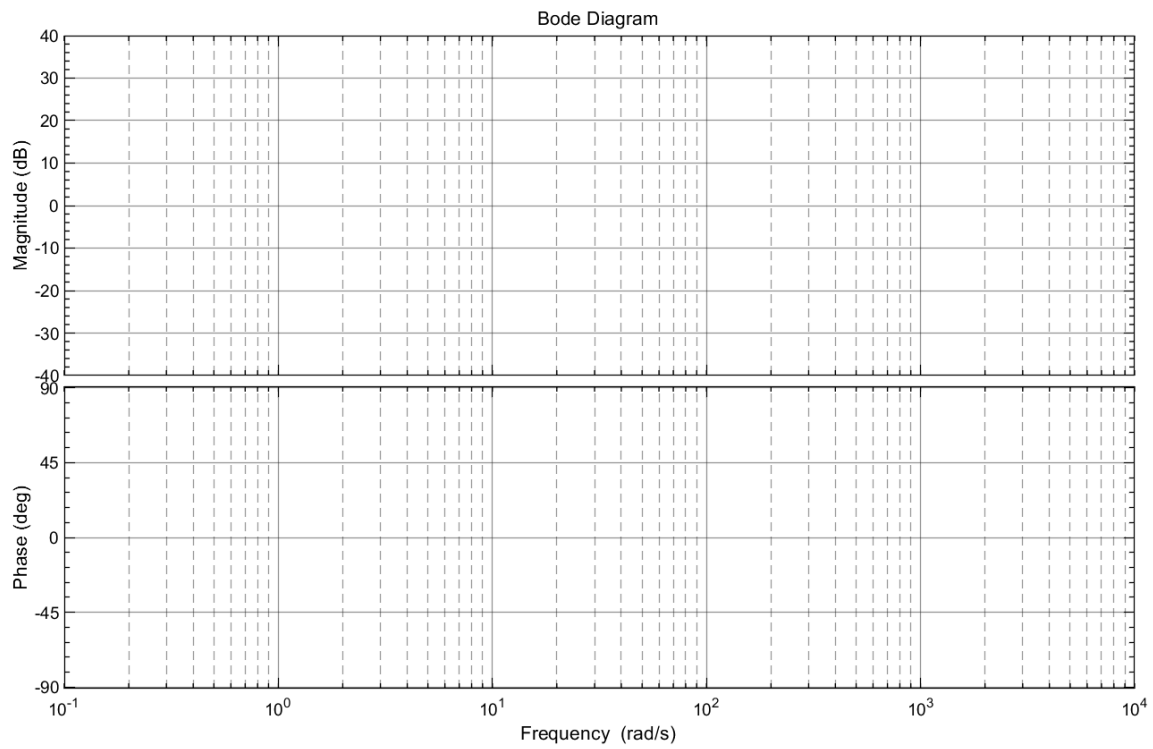
Q2 :



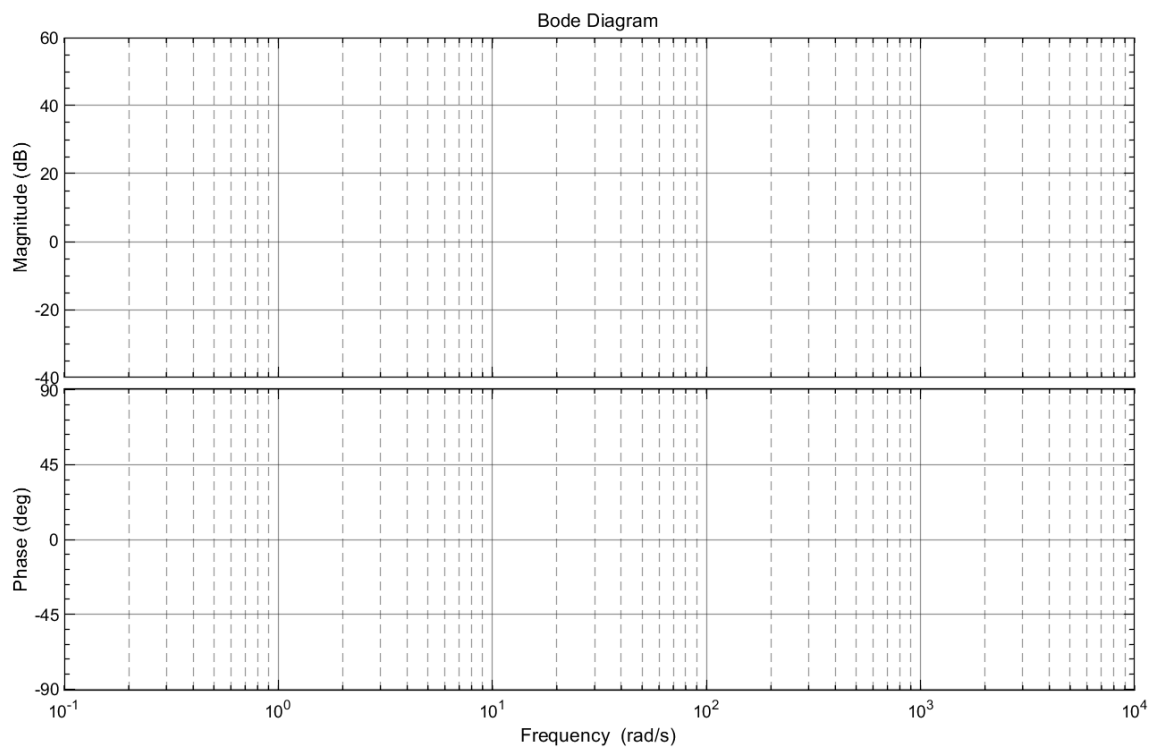
Q3 :



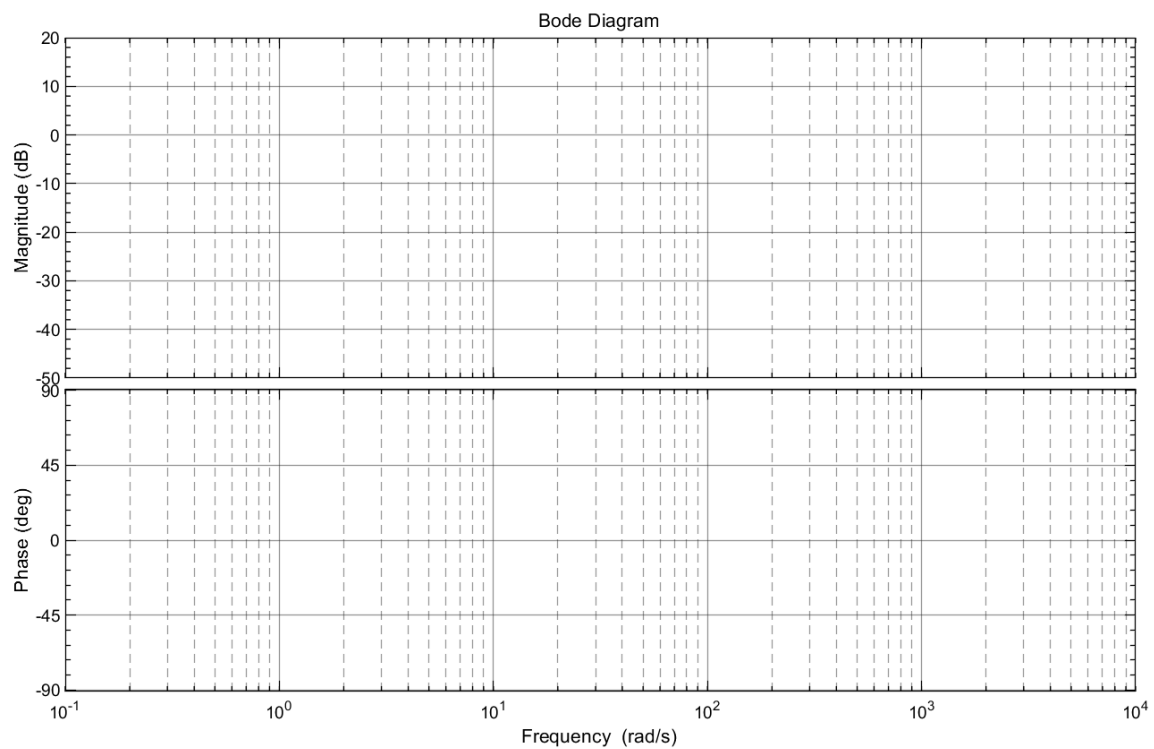
Q4 :



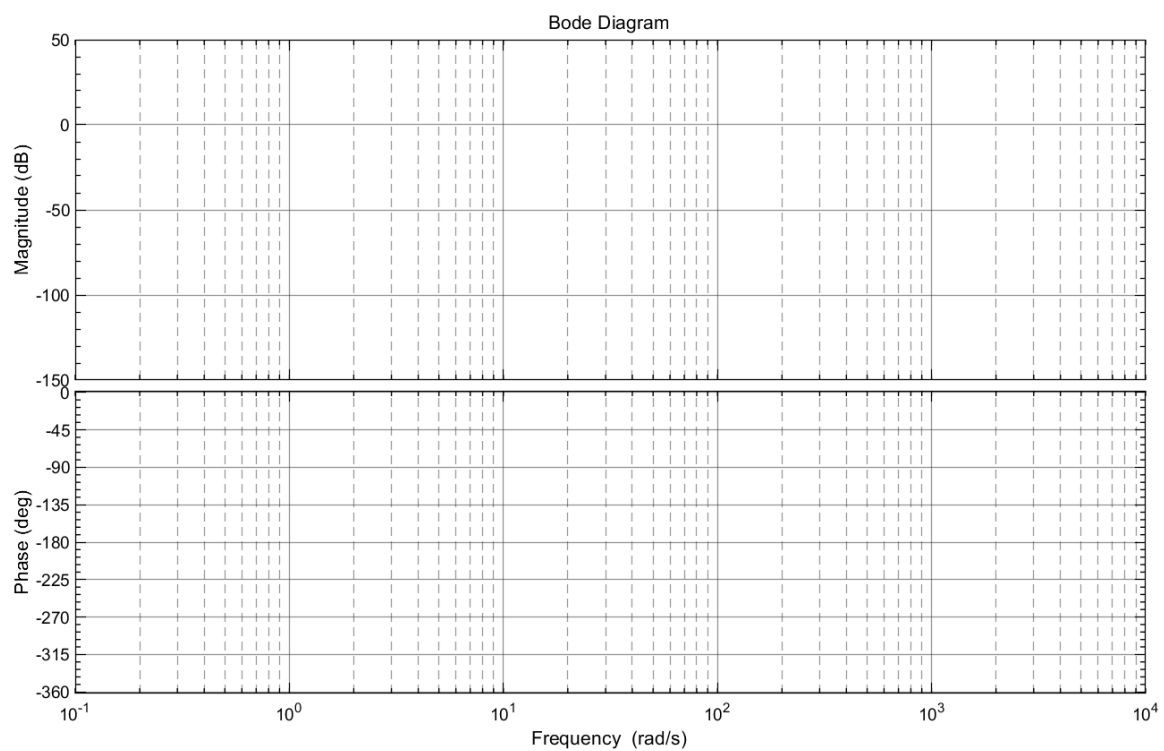
Q5 :



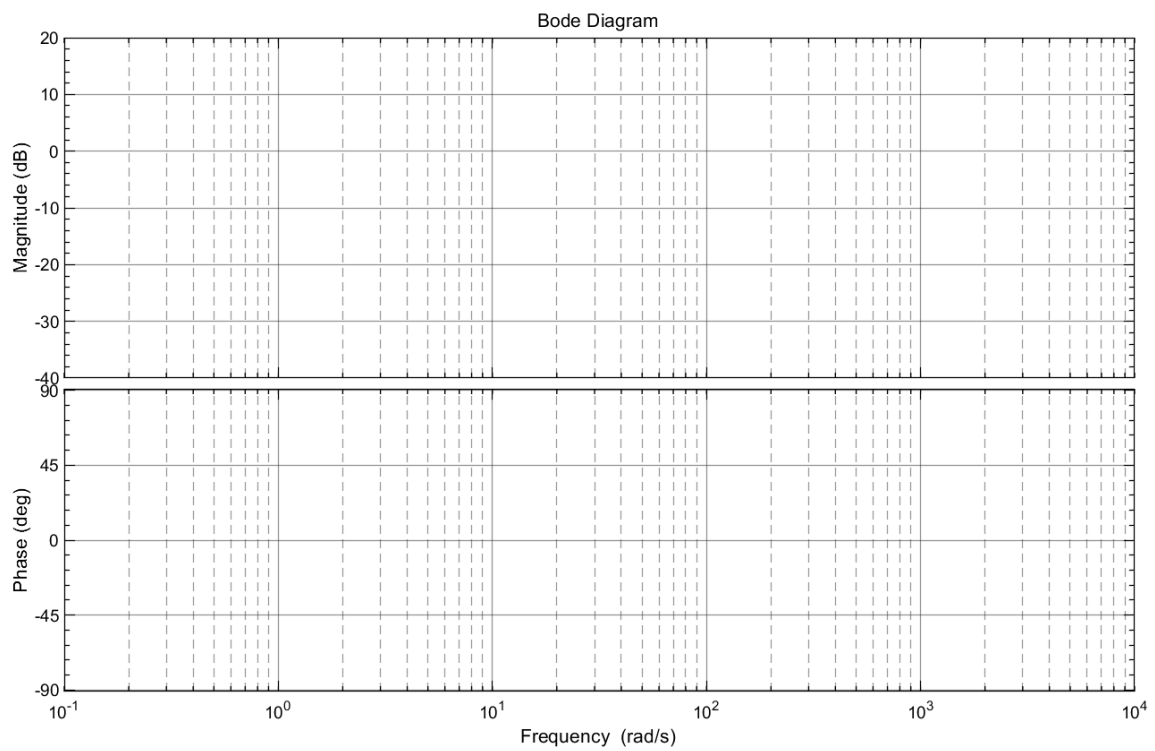
Q6 :



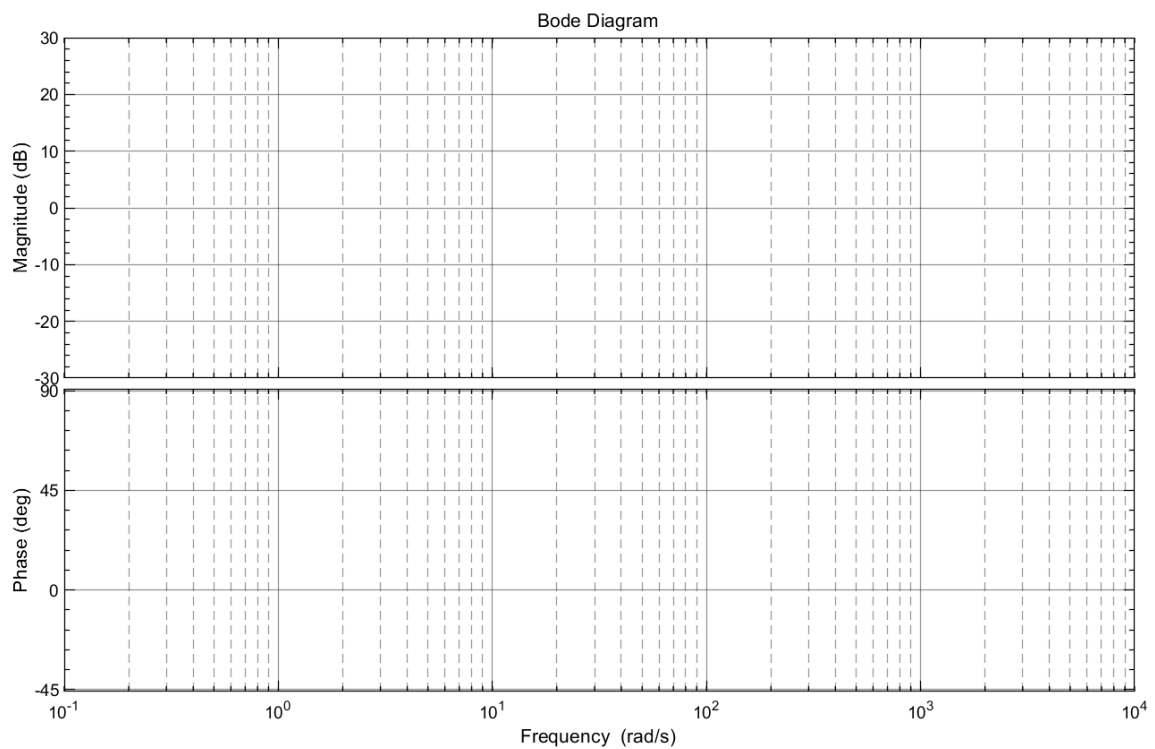
Q7 :



Q8 :



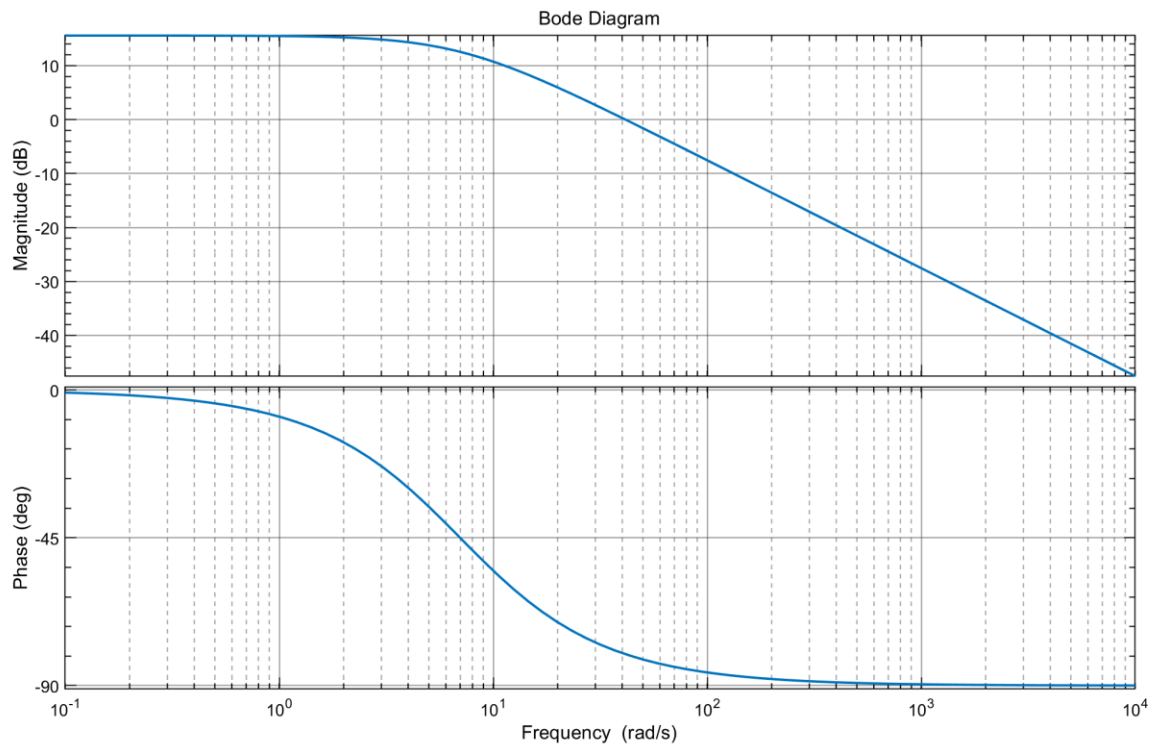
Q9 :



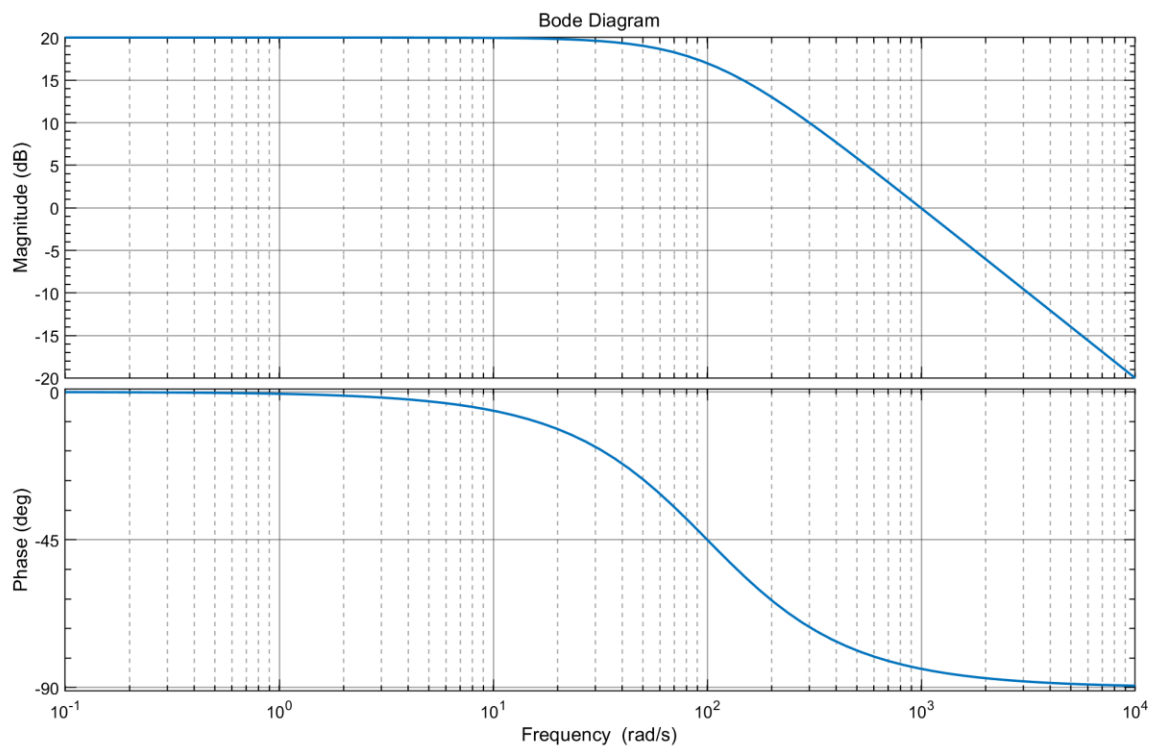
Exercice 3.2 : Détermination de fonction de transfert

Déterminer les fonctions de transferts associées aux diagrammes suivants :

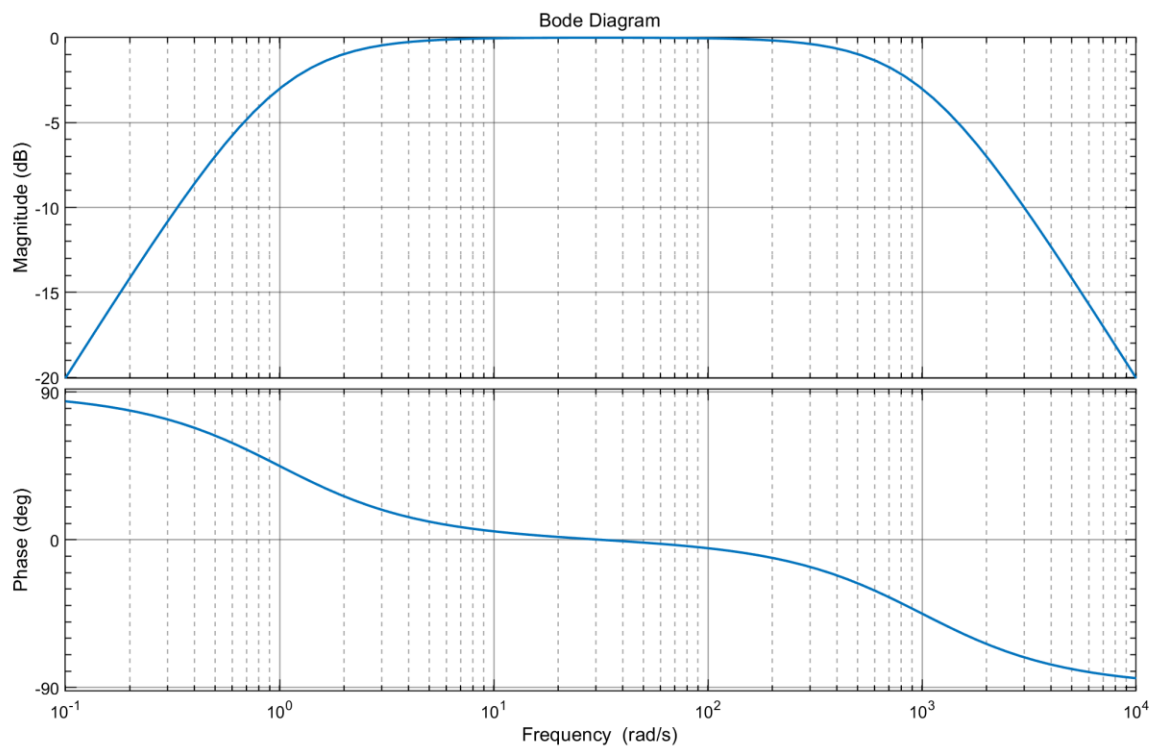
Q1 :



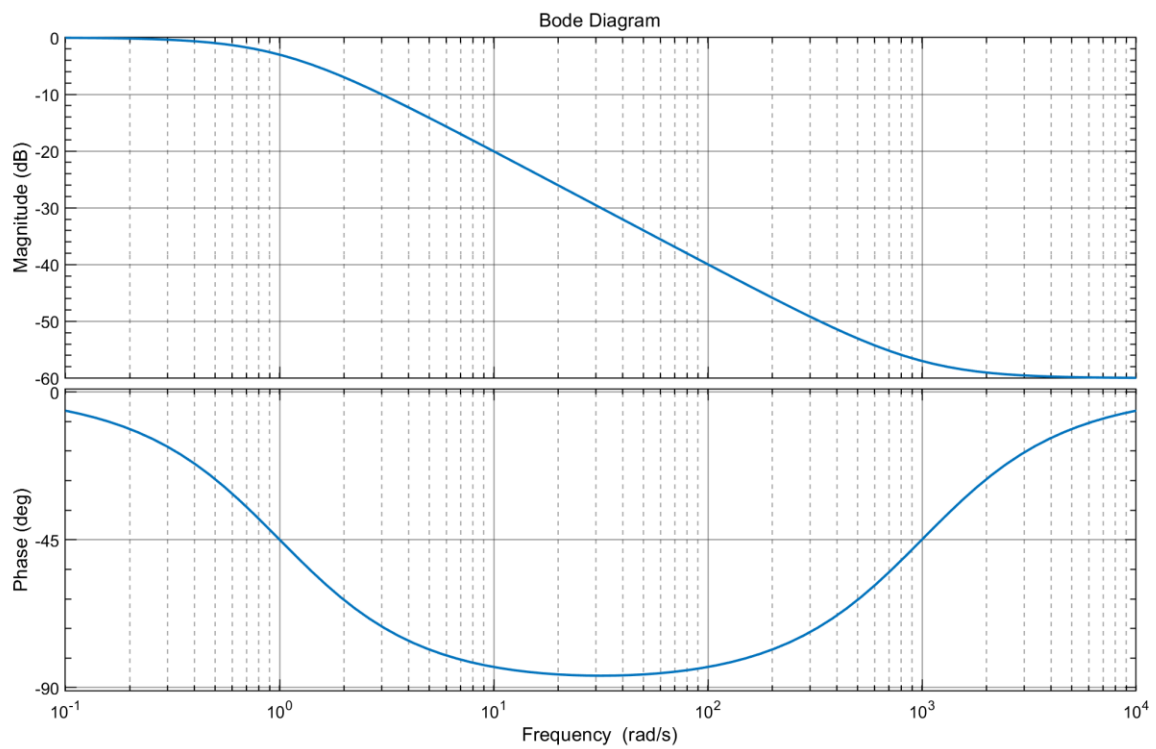
Q2 :



Q3 :



Q4 :



Q5 :

